

Minheng Chen

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EDUCATION

University of Texas at Arlington

Arlington, TX, USA

- Ph.D. in Computer Science

08/2024-present

Southeast University

Nanjing, China

- B.E. in Computer Science and Technology

09/2020-06/2024

AWARDS and HONORS

- Graduate Dean's Research Assistantship, College of Engineering, UTA 2026
- IEEE/CVF Winter Conference on Applications of Computer Vision 2026 Award Finalists (13/832) 2026
- Outstanding Undergraduate Thesis of Southeast University (top 3%) 2024
- Outstanding Undergraduate Thesis of Jiangsu Province (35 students in SEU each year) 2024

RESEARCH EXPERIENCE

Research Assistant, Lab of Imaging Science and Technology, School of CSE, Southeast University, 11/2021-06/2024

PI: Xudong Bao and Youyong Kong

- Developed a CT/X-ray image registration framework and a software system for robotic navigation in minimally invasive spine surgery, obtained software copyright, and contributed to several research papers. (1 ICASSP, 2 ISBI, 1 WACV, 1 Medical Physics, 1 AIIM).
- Participated in the development of advanced algorithms for spinal localization, segmentation, and recognition in orthopedic surgical navigation.

SELECTED PUBLICATION

11. **Minheng Chen**, Tong Chen, Chao Cao, Jing Zhang, Li Su, Tianming Liu, Dajiang Zhu. "Folding-Fiducial Geodesic Representation for Morphometric Similarity Networks in Neurodegenerative Disease Characterization", IEEE International Symposium on Biomedical Imaging, 2026.
10. **Minheng Chen**, Youyong Kong. "Intraoperative 2D/3D Registration via Spherical Similarity Learning and Differentiable Levenberg-Marquardt Optimization", Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026. **Oral, Round 1 Acceptance, 85/1329.**
9. Yan Zhuang*, **Minheng Chen***, Chao Cao, Tong Chen, Jing Zhang, Xiaowei Yu, Yanjun Lyu, Lu Zhang, Tianming Liu, Dajiang Zhu. "GyratNet Sub-network Partitioning via Differentiable Spectral Modularity Optimization", International Workshop on Machine Learning in Medical Imaging, 2025.
8. **Minheng Chen**, Xiaowei Yu, Jing Zhang, Tong Chen, Chao Cao, Yan Zhuang, Yanjun Lyu, Lu Zhang, Tianming Liu, Dajiang Zhu. "Core-Periphery Principle Guided State Space Model for Functional Connectome Classification", International Conference on Medical Image Computing and Computer-Assisted Intervention, 2025. **Early Acceptance, Top 9%**
7. Tong Chen, **Minheng Chen**, Jing Zhang, Yan Zhuang, Chao Cao, Xiaowei Yu, Yanjun Lyu, Lu Zhang, Li Su, Tianming Liu, Dajiang Zhu. "A Unified Continuous Staging Framework for Alzheimer's Disease and Lewy Body Dementia via Hierarchical Anatomical Features", International Conference on Medical Image Computing and Computer-Assisted Intervention, 2025. **Early Acceptance, Top 9%**
6. Junxian Wu, **Minheng Chen**, Xinyi Ke, Tianwang Xun, Xiaoming Jiang, Hongyu Zhou, Lizhi Shao, Youyong Kong. "Learning Heterogeneous Tissues with Mixture of Experts for Gigapixel Whole Slide Images", Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
5. **Minheng Chen**, Chao Cao, Tong Chen, Yan Zhuang, Jing Zhang, Yanjun Lyu, Xiaowei Yu, Lu Zhang, Tianming Liu, Dajiang Zhu. "Using Structural Similarity and Kolmogorov-Arnold Networks for Anatomical Embedding of Cortical Folding Patterns", IEEE International Symposium on Biomedical Imaging, 2025.
4. **Minheng Chen***, Zhirun Zhang*. "Introducing Learning Rate Adaptation CMA-ES Into Rigid Intraoperative 2D/3D Registration for Spinal Surgery", IEEE International Symposium on Biomedical Imaging, 2025.
3. Tonglong Li, **Minheng Chen**, Mingying Li, Chuanyou Li, Youyong Kong. "Automatic X-ray to CT Registration Using Embedding Reconstruction and Lite Cross-Attention", Medical Physics, 2025.
2. **Minheng Chen**, Zhirun Zhang, Shuheng Gu, Zhangyang Ge, Youyong Kong. "Fully Differentiable Correlation-driven 2D/3D Registration for X-Ray to CT Image Fusion", IEEE International Symposium on Biomedical Imaging, 2024.
1. **Minheng Chen**, Zhirun Zhang, Shuheng Gu, Youyong Kong. "Embedded Feature Similarity Optimization with Specific Parameter Initialization for 2D/3D Medical Image Registration", IEEE International Conference on Acoustics, Speech and Signal Processing, 2024.

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Minheng Chen & Zhirun Zhang, SimpleRegi V1.0, registration number: 2023SR0907053, 03/2023

SERVICES

- **Graduate teaching assistant:** CSE 3302 Programming Languages (24 Fall, 25 Spring), CSE 6389 ADV MM, GRAPHICS, IMAGE PROC (25 Fall), CSE 3380 Linear Algebra for CSE(25 Fall)
- **Conference reviewer:** ISBI 2025-26, AAAI 2026, MIDL 2026, MICCAI 2026, AMIA 2026
- **Journal reviewer:** IEEE TMI, IEEE JBHI

SKILLS

Languages: Chinese (Native); English (Proficient)

Computer Skills: Proficient in C/C++, Python, Java, Matlab, LaTeX, and Office, familiar with PyTorch framework